

Customer Churn EDA Report

1. Introduction

Customer churn means when users stop using a service. It is important because losing customers affects business revenue. By analyzing churn, companies can understand why customers leave and take steps to retain them.

2. Dataset Details

This dataset contains customer information from a streaming platform. It includes details like age, subscription type, watch behavior, and payment method.

- Number of columns: 14
 - Target variable: **churned** (whether a customer left or not)
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3. Data Cleaning Summary

- No major missing values were found in the dataset.
 - No duplicate records were observed..
 - Drop the customer id column
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4. Exploratory Data Analysis

- Count plots and pie charts showed the distribution of churned vs non-churned users.
 - Univariate analysis and Bivariate analysis helps to understand the columns more deeply.
 - Boxplots were used to detect outliers.
 - Bar charts and countplots showed how churn varies across categories like subscription type and device.
 - Heatmap showed relationships between numerical features.
 - Scatter plots helped analyze relationships like watch hours vs monthly fee.
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5. Key Findings

- Users with low watch time and high inactivity are more likely to churn.
- Higher monthly fees increase churn, especially for low-usage users.

- Basic subscription users churn more than premium users.
 - Users with more profiles (family usage) churn less.
 - Low engagement + high cost users are at highest risk.
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6. Business Recommendations

- Improve user engagement with personalized recommendations.
 - Target inactive users with offers or reminders.
 - Provide flexible pricing plans.
 - Encourage family subscriptions.
 - Improve app experience across devices.
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7. Conclusion

The analysis shows that churn mainly depends on user engagement, pricing, and subscription type. By improving user experience and targeting high-risk users, companies can reduce churn and increase customer retention.